==========

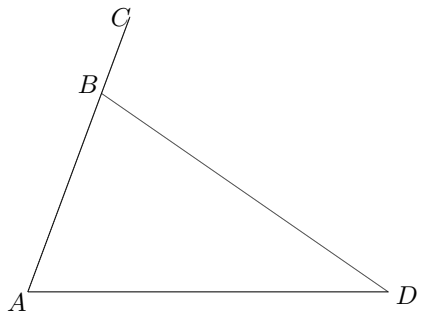
NMC 2016 PRIMARY LEVEL PAPER 1 QUESTIONS
SECTION A

Qn1. Two vertical poles, one 10 metres high and the other 15 metres high, stand 12 metres apart. Find the distance, in metres, between the tops of the poles.

Qn2. Jane is standing on a staircase. She walks down 4 steps, up 3 steps, down 6 steps, up 2 steps, up 9 steps, and down 2 steps. Where is she standing in relation to the step on which she started?

Qn3. What is the sum of the next two numbers in the sequence: 6, 11, 21, 36, 56, ..., ...?

Qn4. In the figure, angle $BAD = 60$ degrees, angle $CBD = 85$ degrees. Angle $ADB =$ ___ degrees.



Qn5. Alice needs one hour to do a certain job. Her older sister Annet can do the same job in 30 minutes. How many minutes will it take them to do the job if they work together at the given rates?

Qn6. The rectangular dinning table at school is 3 times as long as it is wide. If its length is reduced by 3m and its width is increased by 3m, it would become a square. What is the area of the dinning table?

Qn7. The ages of Agnes, Ben and Carol are each prime numbers. Agnes is the youngest. The sum of the ages of Agnes and Ben is equal to Carol's age. How old is Agnes?

Qn8. Arrange the digits 1, 1, 2, 2, 3, 3 as a six-digit number in which the 1's are separated by one digit, the 2's are separated by two digits, and the 3's are separated by three digits.

Qn9. If South-east becomes North, North-east becomes West and so on, what will West become?

Qn10. A school Library has an average of 510 visitors on Sundays and 240 on other days. What is the average number of visitors per day in a month of 30 days (assume the 1st day of that month falls on a Sunday)?

SECTION B

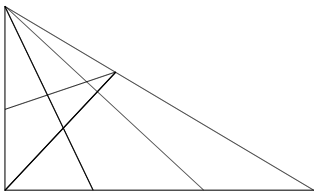
Qn11. Annet arranges all the counting numbers in the 10 columns as shown below:

A	B	C	D	E	F	G	H	I	J
1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	..	..	..	..	..	..

(a) Under what letter (column) will the number 1000 appear? Explain.

(b) If there were 6 columns instead of 10, under what column would the number 1000 have appeared?

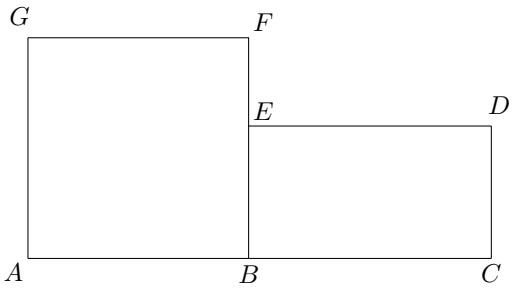
Qn12. Count all the triangles in the figure. How many are they?



Qn13. The age of John's father is the same as the age of his mother with digits reversed. The sum of their ages is 99 and the father is 9 years older than the mother. How old is the father?

Qn14. **STEP** 1: ADD 4
STEP 2: SUBTRACT 1
STEP 3: IF ≤ 15, JUMP TO STEP 1 AND CONTINUE FROM THERE; OTHERWISE, PROCEED TO STEP 4
STEP 4: ADD 3
STEP 5: IF ≥ 18, SUBTRACT 2
If you start with 2 and then apply the above instructions as given in the steps, what is the end result?

Qn15. In the figure, $ABFG$ is a square, $BCDE$ is a rectangle. The length of CD is 25 cm. The length of BC is the same as the length of AB. The perimeter of the entire figure is 180 cm.



(a) What is the length of FE?

(b) What is the area of the figure?

NMC 2016 PRIMARY LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. Distance between the poles is $12m$ and the vertical height between the tops of the poles is $15 - 10 = 5m$. By Pythagoras theorem, the direct distance between the tops of the poles is
$\sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$metres.

Qn2. Jane is standing $-4 + 3 - 6 + 2 + 9 - 2 = +2$ in relation to the step on which she started.

We can alternatively use a number line to arrive at the same solution.

Qn3. The sequence $6, 11, 21, 36, 56, \cdots$ is increasing respectively by adding $5, 10, 15, 20$. Therefore the next two numbers will be obtained by adding 25 to 56 and then $25 + 56 + 30$. The numbers are 81 and 111.

Qn4.

$$60^\circ + x = 85^\circ \Longrightarrow x = 25^\circ.$$

Qn5. Let the job be j. Then the rate of working $= \frac{job}{time}$.

For Alice, the rate of work $= \frac{j}{60}$work/min and for sister Annet, the rate of work $= \frac{j}{30}$work/min.

The total rate of work for both sisters $=\frac{j}{60}+\frac{j}{30}=\frac{j+2j}{60}=\frac{j}{20}$work/min.
Therefore the required time to complete the job by both sisters is

$$\frac{Total\ work\ done}{rate}=\frac{j}{\frac{j}{20}}=j\times\frac{20}{j}=20\text{min.}$$

Qn6. Let x be the width of the rectangular dinning table. Then its length is $3x$. Reducing the length by 3 gives a new length of $3x-3$ and increasing its width by 3 gives a new width of $x+3$. The result is a square implying that $3x-3=x+3\Longrightarrow x=3$. The area of the dinning table is thus $x\times 3x=3x^2=3\times 3^2=3\times 9=27\text{m}^2$.

Qn7. Let the respective ages be A, B and C. Then from the information provided, we have that $A<B$, $A<C$ and $A+B=C$. However A, B and C are prime numbers and the condition $A<C$ implies C is an odd number since there is only one even prime number which is 2. Since C is odd and $A+B=C$ then one of A and B must be even and the other odd. But there is only one even prime number which is 2. Hence $A=2$ since $A<B$ and $A<C$. Therefore Agnes is 2 years old.

Qn8. The six digit numbers are 231213 or 312132.

Qn9. South-east becomes North through a turning of 135°. Similarly North-east turns by the same angle to $West$. Thus West will turn by the same angle of 135° to South-east.

Qn10. If $1st$ day of 30 days month is on Sunday, then the month will have 5 sundays ie (1st, 8th, 15th, 22nd, 29th). The remaining 25 days are not sunday days.
Number of visitors for Sunday days $=510\times 5=2550$.
Number of visitors for the remaining 25 days $=240\times 25=6000$.
The total number of visitors in the month is $2550+6000=8550$.
Therefore the average number of visitors per day $=\frac{8550}{30}=285$.

SECTION B

Qn11. (a) There are 10 columns. Column J consists of numbers which are multiples of 10. Since 1000 is a multiple of 10, it is in this column J that 1000 appears and specifically in the $\frac{1000}{10}=100$th row.

(b) $\frac{1000}{6}$ gives 166 and leaves a remainder of 4. Since there were only 6 columns, $6\times 166=996$ appears in 6th column. Therefore 997 appears in the 1st column, 998 appears in the 2nd column, 999 appears in the 3rd column and 1000 appears in the 4th column which is column D.

Qn12. The triangles in the figure are $ABH, AHI, AIG, ABI, AHG, ABG, ACJ, AJK, AKG, ACK, AGJ$
$ACG, ACD, ADE, AEF, ACE, ADF, ACF, IKG, HJG, BGC, CJD, CKE, CGF$

There are 24 triangles.

Qn13. We can write a two digit number xy as $10x+y$ ie x tens and y ones. Let the age of John's father be xy and his mother be yx. Then

$$\begin{aligned} xy+yx &= 99 \\ 10x+y+10y+x &= 99 \quad\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1) \\ \text{Further more} & \\ xy-yx &= 9 \\ 10x+y-(10y+x) &= 9 \quad\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2) \end{aligned}$$

Adding the two equations $(1)+(2)$ gives $10x+y=54\Longrightarrow xy=54$. Therefore age of John's father is 54.

Qn14. If we start with 2 and then apply the instructions as given in steps 1 to 5 we have the following.

STEP	
1	$2+4=6$
2	$6-1=5<15$
3	$5+4=9$
2	$9-1=8<15$
3	$8+4=12$
2	$12-1=11<15$
3	$11+4=15$
2	$15-1=14<15$
3	$14+4=18$
2	$18-1=17>15$
4	$17+3=20>18$
5	$20-2=18$

Therefore the card result is 18.

Qn15. (a) Let the length of $AB=AC=CF=x$.
Perimeter of the figure is $x+x+x+x+25+x+x-25=180 \Longrightarrow 6x=180 \Longrightarrow x=30$. Therefore the length $FE=x-25=30-25=5$ cm.

(b) Area of the figure is $x^2\times(x\times 25)=30^2+(30\times 25)=900+750=1650\text{cm}^2$.

NMC 2016 PRIMARY LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. In the text line of numbers below, insert "+" between the numbers as many times as necessary so that 987654321 = 90.

Qn2. Evaluate

$$\frac{3^3+3\times 3+3}{3^2+3^1+3^0}$$

Qn3. What is the value of $\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}$?

Qn4. At a farm, one plant is currently 1 meter tall and grows at a rate of 6cm per week. A second plant is currently 0.7 meters tall and grows at a rate of 9cm per week. In how many weeks will both plants be of the same height?

Qn5. Kahwa thinks of a whole number less than 30. When he divides the number by 6 the remainder is 3. When he divides the number by 8 the remainder is 5. What is the number?

Qn6. A sale advert read: "Buy three mathematics books at the regular price and get the fourth book at Shs. 3000." Kato paid Shs. 24000 for the set of four books at the sale. What was the regular price of one book?

Qn7. It is 215 minutes to midday. What time is it on a 24-hour digital clock?

Qn8. Each game in a knock-out tournament involves 7 players and only the winner of the game remains in the tournament (the other players are knocked out). The tournament was played in 8 games to declare the final winner. How many players were there at the start of the tournament?

Qn9. Babirye earns Shs. $300,000$ per week for a 44 hour week. Her weekly salary is increased by 8% and her hours are reduced by a fifth. How much does she earn now per hour?

Qn10. In the figure shown, the overlap of rectangles ABCD and EFGH is also a rectangle. The area of ABCD is $21cm^2$. The area of EFGH is $50cm^2$. $AB=3cm$ and $EH=5cm$. Find the area of the entire figure.